

Exam policy:

- *The exam is 1 hour (7:30–8:30 PM) and is closed book, closed notes.*
- *You are allowed one hand-written cheat sheet of at most one two-sided page. Please hand in your cheat sheet with your solutions.*
- *Justify all your answers. It is not enough to just have the right answer. You must also convince us that you got the right answer for the right reasons.*
- *No calculators.*

Each of the three questions is worth an equal number of points.

The exam questions are on the other side of this page.

Please do not turn over this page until you are told to start the exam.

Good luck!

Q. 1 (Twitter). Your twitter feed is a hive of activity. You are a loyal follower of Donald Trump (@realDonaldTrump), Justin Bieber (@justinbieber), and of Oprah Winfrey (@Oprah). Trump tweets on average once every 15 min., Bieber on average once every 5 min., and Oprah once per hour, all independently according to Poisson processes.

- What is the probability that a tweet from Trump arrives before Bieber has tweeted five times?
- What is the expected time until the first tweet, conditionally on the information that the first tweet you see is from Trump?
- What is the expected amount of time it takes before you have seen at least one tweet from Bieber and Oprah?

Q. 2 (Smoking is bad for you). Two men lead independent lives. One is a non-smoker, and his hazard rate function is h . The other is a smoker, and his hazard rate function is $2h$. We do not assume the hazard rate function is constant.

- What is the density of the lifetime of the nonsmoker?
- What is the probability that the smoker outlasts the nonsmoker?
[Give an explicit numerical answer.]

Q. 3 (An asymmetric game). You play the following game at the casino. In every round, a fair coin (heads or tails with equal probability) is flipped independently. If it comes up heads, you gain a dollar. If it comes up tails, you lose two dollars. You start with 2 dollars. The game ends when you have either lost all your money (your wealth is 0 dollars or less), or when your wealth has reached 4 dollars.

- Does the game end eventually?
- What is the expected time at which the game ends?
- What is the probability that the game ends with you being in debt (wealth < 0)?